AI-Based ER Diagram Generation Using GitHub Copilot

Prompt for Github Copilot-

Create a complete Entity-Relationship (ER) diagram for an Online Library ManagementSystem.

ENTITIES AND ATTRIBUTES:

1. STUDENT
- student_id (Primary Key)
- name
- email

2. RESERVATION
- reservation_id (Primary Key)
- reservation_date
- status
- student_id (Foreign Key)
- book_id (Foreign Key)

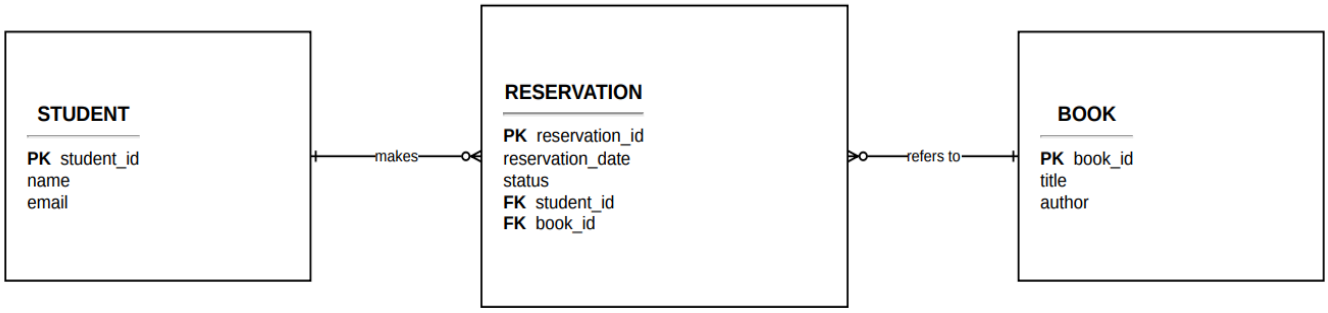
3. BOOK
- book_id (Primary Key)
- title
- author

RELATIONSHIPS:

1. One STUDENT can make zero or many RESERVATIONS.
   Each RESERVATION belongs to exactly one STUDENT.
2. One BOOK can appear in zero or many RESERVATIONS over time.
   Each RESERVATION refers to exactly one BOOK.

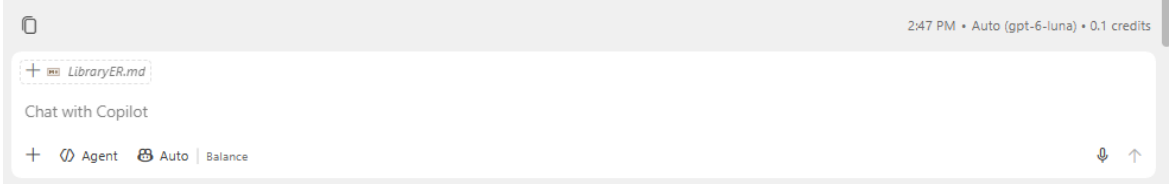
TECHNICAL REQUIREMENTS:
- Generate complete, valid draw.io XML that can be imported directly into diagrams.net.
- Use proper rectangular entity boxes with clearly displayed attributes.
- Display entity names prominently.
- Mark primary keys as PK and foreign keys as FK.
- Use correct crow's-foot notation for one-to-many relationships.
- Ensure STUDENT has the one end and RESERVATION has the many end.
- Ensure BOOK has the one end and RESERVATION has the many end.
- Arrange STUDENT on the left, RESERVATION in the center, and BOOK on the right.
- Use straight, properly connected relationship lines.
- Add relationship labels: "makes" and "refers to".
- Use a white page background with black text, borders, and connectors.
- Use a clean, professional layout with consistent font sizes and spacing.
- Keep all entities and relationships within a single landscape page.
- Make the diagram suitable for printing in a college practical manual.
- Avoid overlapping text, connectors, and entity boxes.
- Use a standard draw.io XML structure with valid cell IDs and geometry.
- Return only the complete XML, without explanations or Markdown code fences.

ER Diagram of Online Library Management System -



XML Code Generated Using GitHub Copilot -


```
<mxfile host="app.diagrams.net" modified="2026-10-15T09:16:00.000Z" agent="Copilot" version="24.7.17" type="device"> <diagram id="online-library-er" name="Online Library ER Diagram"> <mxGraphModel dx="1200" dy="650" grid="1" gridSize="10" guides="1" tooltips="1" connect="1" arrows="1" fold="1" page="1" pageScale="1" pageWidth="1200" pageHeight="650" background="#ffffff" math="0" shadow="0"> <root> <mxCell id="0"/> <mxCell id="1" parent="0"/> <mxCell id="student" value="&lt;div style=&quot;text-align:center;font-size:16px&quot;&gt;&lt;b&gt;STUDENT&lt;/b&gt;&lt;/div&gt;&lt;hr&gt;&lt;div&gt;&lt;b&gt;PK&lt;/b&gt;&amp;nbsp; student_id&lt;/div&gt;&lt;div&gt;name&lt;/div&gt;&lt;div&gt;email&lt;/div&gt;" style="rounded=0;whiteSpace=wrap;html=1;align=left;verticalAlign=middle;spacingLeft=16;fontSize=14;strokeColor=#000000;fillColor=#ffffff;fontColor=#000000;strokeWidth=1.5;" vertex="1" parent="1"> <mxGeometry x="60" y="220" width="250" height="190" as="geometry"/> </mxCell> <mxCell id="reservation" value="&lt;div style=&quot;text-align:center;font-size:16px&quot;&gt;&lt;b&gt;RESERVATION&lt;/b&gt;&lt;/div&gt;&lt;hr&gt;&lt;div&gt;&lt;b&gt;PK&lt;/b&gt;&amp;nbsp; reservation_id&lt;/div&gt;&lt;div&gt;reservation_date&lt;/div&gt;&lt;div&gt;status&lt;/div&gt;&lt;div&gt;&lt;b&gt;FK&lt;/b&gt;&amp;nbsp; student_id&lt;/div&gt;&lt;div&gt;&lt;b&gt;FK&lt;/b&gt;&amp;nbsp; book_id&lt;/div&gt;" style="rounded=0;whiteSpace=wrap;html=1;align=left;verticalAlign=middle;spacingLeft=16;fontSize=14;strokeColor=#000000;fillColor=#ffffff;fontColor=#000000;strokeWidth=1.5;" vertex="1" parent="1"> <mxGeometry x="450" y="200" width="300" height="230" as="geometry"/> </mxCell> <mxCell id="book" value="&lt;div style=&quot;text-align:center;font-size:16px&quot;&gt;&lt;b&gt;BOOK&lt;/b&gt;&lt;/div&gt;&lt;hr&gt;&lt;div&gt;&lt;b&gt;PK&lt;/b&gt;&amp;nbsp; book_id&lt;/div&gt;&lt;div&gt;title&lt;/div&gt;&lt;div&gt;author&lt;/div&gt;" style="rounded=0;whiteSpace=wrap;html=1;align=left;verticalAlign=middle;spacingLeft=16;fontSize=14;strokeColor=#000000;fillColor=#ffffff;fontColor=#000000;strokeWidth=1.5;" vertex="1" parent="1"> <mxGeometry x="890" y="220" width="250" height="190" as="geometry"/> </mxCell> <mxCell id="student-reservation" value="makes" style="edgeStyle=none;rounded=0;html=1;strokeColor=#000000;fontColor=#000000;strokeWidth=1.5;startArrow=ERone;startFill=0;endArrow=ERzeroToMany;endFill=0;fontSize=12;labelBackgroundColor=#ffffff;exitX=1;exitY=0.5;entryX=0;entryY=0.5;" edge="1" parent="1" source="student" target="reservation"> <mxGeometry relative="1" as="geometry"/> </mxCell> <mxCell id="reservation-book" value="refers to" style="edgeStyle=none;rounded=0;html=1;strokeColor=#000000;fontColor=#000000;strokeWidth=1.5;startArrow=ERzeroToMany;startFill=0;endArrow=ERone;endFill=0;fontSize=12;labelBackgroundColor=#ffffff;exitX=1;exitY=0.5;entryX=0;entryY=0.5;" edge="1" parent="1" source="reservation" target="book"> <mxGeometry relative="1" as="geometry"/> </mxCell> </root> </mxGraphModel> </diagram> </mxfile>
```

Description-

The generated XML was imported into draw.io, and the ER diagram was rendered successfully. The diagram represents the entities, their attributes, and the one-to-many relationships between STUDENT and RESERVATION, and between BOOK and RESERVATION.